CHAPTER

1 MATHEMATICAL BACKGROUND

1.1 PREFACE

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Testing some equations: let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\boldsymbol{c} = \alpha\boldsymbol{x} + \beta\boldsymbol{y} \in \mathbb{R}^n. \tag{1.1}$$

However, if $\boldsymbol{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha\boldsymbol{x} \in \mathbb{R}^m$, and therefore $\boldsymbol{c}$ is not defined (i.e. Equation 1.1 is irrelevant).

Note 1.1 Test note

This is a test note. I will try to also refernce it later.

!

Now, I wish to refernce 1.1. Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = \begin{bmatrix}1, & 2, & -3\end{bmatrix}$ and $y = \begin{bmatrix}5, & -1\end{bmatrix}$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha\boldsymbol{x} \in \mathbb{R}^3$, while $\beta\boldsymbol{y} \in \mathbb{R}^2$. This means that $\alpha\boldsymbol{x} + \beta\boldsymbol{y}$ is not defined.

I believe it is now time for some figures! See Figure 1.1.

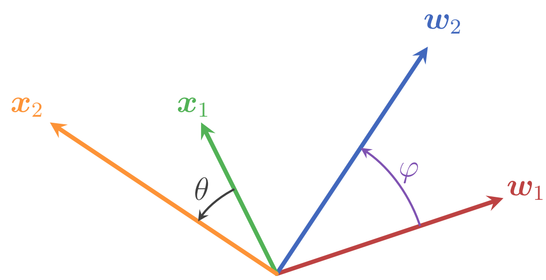

Figure 1.1: This is the first test figure. Does it look ok?

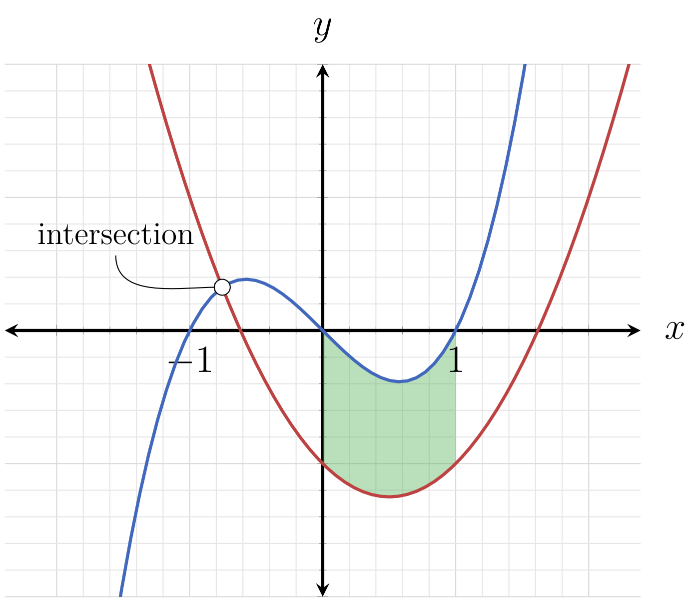

Figure 1.2: And this is another test figure! I hope it works.

CHAPTER

THE BASICS OF LINEAR ALGEBRA

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CHAPTER

3 BIBLIOGRAPHY

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